

LECTURE-16

Friday.
27/4/26

I Discrete Prob. Distribution:

If we need prob. of any event from 3 coins tossed; we can make sample space of 8 possibilities & find relative freq. of either 2 heads, 3 tails etc. BUT if we've suppose 100 coins; we can't make ~~its~~ sample space & keep going to find relative freq. for just a single event.
 $\therefore$ for that we'll use discrete prob. distribution.

⇒ Bernoulli Trials: (Experiment)

- 1) It consists of multiple repeated trials
- 2) Results are either 'success' (p) or 'failure' (q).
- 3) Prob. of success = p , prob of failure = q .
- 4) Trials are independent.

e.g.: Coin toss is repeated trials; $p=0.5$, $q=0.5$; Each toss is independent ✓

• A programmer solving a question again & again ISN'T Bernoulli trial as each time he gets better at his skill $\therefore p$ 'changes' hence it does not remain independent.

⇒ Binomial Distribution:

r = No. of times of specific event we wanna find prob. of.

p = S(success), q = F(fail)

r	Sample Space	Formula
0	FFF	$p^0 \cdot q^3 \cdot \frac{3!}{(3! \cdot 0!)}$
1	SFF	$p^1 \cdot q^2 \cdot \frac{3!}{1! \cdot 2!}$
2	SF	$p^2 \cdot q^1 \cdot \frac{3!}{2! \cdot 1!}$
$\vdots$		
n		$p^r \cdot q^{n-r} \cdot \frac{n!}{(n-r)! \cdot r!}$

No. of ~~FFF~~ combinations of
→ SFF. e.g. could be FSF, FFS...

} Formula of Binomial Distribution:

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r}$$

Mean = np

Variance = npq

Shape ⇒

$p=1/2 \rightarrow$ Symmetric

$p > 0.5 \rightarrow$ Negatively skewed

$p < 0.5 \rightarrow$ Positively skewed.

$$n=0, 1, 2, \dots, n.$$

' n ' and ' p ' are parameters of $P(X=r)$.

NOTE:

All four properties of Bernoulli Trials (above) are also properties of Binomial distri. along with a 5th property i.e.

5) Experiment is repeated fixed number of ' n ' times.

Prove it's a Binomial distr:

- 2 possible outcomes; A wins or loses
- $p = \text{success} = 2/3$ (given)
- Each game is Independent to e/o
- Repeated 8 games $\therefore n=8$.

Q: A & B plays a game. Prob. of A winning is $2/3$.
In series of 8 matches, find prob. of A winning:

(a) Exactly 4 games. $n=8, r=4, p=2/3, q=1/3$

$$P(X=4) = {}^8C_4 \cdot (2/3)^4 \cdot (1/3)^{8-4}$$

$$= \frac{1120}{6561} = \boxed{0.1707}$$

(b) at least 4 games:

$$P(X \geq 4) = \sum_{r=4}^8 ({}^8C_r \cdot (2/3)^r \cdot (1/3)^{8-r})$$

$$= [{}^8C_4 \cdot (2/3)^4 \cdot (1/3)^4] + [{}^8C_5 \cdot (2/3)^5 \cdot (1/3)^3] + \dots + [{}^8C_8 \cdot (2/3)^8 \cdot (1/3)^0]$$

$$= \boxed{0.9121}$$

(c) 3 to 6 games: $P(3 \leq r \leq 6) = \sum_{r=3}^6 ({}^8C_r \cdot (2/3)^r \cdot (1/3)^{8-r})$

(d) Mean, variance & SD = ?

Mean $\Rightarrow p \cdot n = 8 \cdot (2/3) = \boxed{16/3}$; Variance $\Rightarrow npq = 8(2/3)(1/3) = \boxed{16/9}$;

SD $\Rightarrow \sqrt{\text{variance}} = \sqrt{16/9} = \boxed{4/3}$

(e) Shape. $n=8, p=0.666 (2/3) \rightarrow$ Negative skewed as: $p > 1/2$

